

# MACS 40550: Final Project Draft

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## Research Question

Under what conditions does social observation transform initially indifferent agents into active participants in scarcity purchasing, and when does the cascade of rising demand collapse before supply is exhausted?

## Literature Review and Framing

The relationship between scarcity and perceived value is well-established in the literature. Worchel et al. (1975) demonstrated in a classic experiment that limiting the availability of an object increases its perceived desirability independently of any change in its intrinsic quality, the finding since replicated and extended across numerous consumer contexts. Lynn (1991) conducted a quantitative review of the commodity theory literature and confirmed that scarcity effects on value are robust across product types and cultural settings. These findings establish that limited availability is not merely a constraint on supply but an active signal that shapes how consumers evaluate goods.

The level of consumer demand itself are that mostly treated as fix in scarcity literature, the existing computational models of consumer behavior tend to treat demand as exogenous, assuming that the number of people who want a product is determined prior to and independently of the social process by which purchasing unfolds (Janssen and Jager, 2001). This assumption becomes problematic when we consider platforms and environments in which consumers can observe each other's interest and behavior in real time, such as online limited-edition, live-streamed product launches, or group purchasing communities. In these settings, demand is socially constructed rather than individually fixed.

There are two literatures directly to the possibility. Granovetter (1978) The threshold model of collective behavior provides a framework for understanding how individual-level heterogeneity in decision thresholds can produce aggregate patterns of adoption or non-adoption that are highly sensitive to initial conditions. In the model, each individual has a threshold representing the number of others who must act before

they are willing to act themselves. The resulting dynamics produce tipping points and cascade failures that are not predictable from the distribution of individual preferences alone. Bikhchandani et al. (1992) shows the complementary mechanism through informational cascade theory, showing that rational individuals may choose to disregard their own private information and imitate others when the observed behavior of predecessors is sufficiently informative. Together, these frameworks predict that social observation can amplify or suppress demand in ways that are highly nonlinear and path-dependent.

The gap that I want to fill is a model in which demand itself as an emergent outcome rather than the starting condition, shaped by the weighted influence of neighbors whose interest is observable and tie strength varies. While existing models tend to isolate one dimension of this process at a time, the model integrates scarcity, social contagion, and network tie strength within a single framework to examine how demand emerges from their interaction.

## Mechanism Description

The underlying social mechanism I model is one of demand contagion moderated by scarcity. Each agent begins with their own private interest level reflect their baseline disposition toward the product, which may be low or high depending on individual heterogeneity. As the agent observe neighbors expressing their interest or actively purchasing, they update their own interest upward through a social learning process. Conversely, neighbors who visibility fail to obtain the product send a discouraging signal that suppresses their interest. When an agent's interest level crosses their personal threshold, they join the purchasing queue.

The critical feature of this mechanism is that the tie strength between agent is not uniform. The connection to close friends and family carries full influence, while connections to strangers carry substantially reduced influence, following the strong/weak tie distinction established in social network theory. This means that the same level of aggregate network activity will have different effects depending on whether it is observed through strong or weak ties; it allows the model to capture the intuition that peer behavior matters more when the peers are people we trust.

Scarcity is not the fixed parameter; it is the dynamic constraint that interacts with the cascade. As agents successfully purchase and supply decreases, later-arriving agents who have been influenced to act may find supply exhausted and generate negative feedback that can arrest or reverse the cascade. Whether the cascade reaches most of the population or collapses prematurely depends on the interplay between the speed of social contagion and the rate at which supply is depleted, which is in turn mediated by restocking. The tipping-point structure is the core aggregate outcome the model is designed to produce and explore.

## Description of Model

The model is a network-based spatial structure. Agents are placed on nodes of either Erdős–Rényi random graph or a Watts-Strogatz small-world graph, which is select by users. The graph of Erdős–Rényi creates connections with uniform probability, producing a network without strong local clustering. The Watts-Strogatz graph starts with a ring lattice in which each node is connected to its nearest neighbors and then rewires a proportion of edges at random to produce a locally clustered network with some long-range shortcuts that approximates the structure of real social networks more closely.

Each agent is initialized with two key attributes. The first is an interest level from Gaussian distribution where most agent begin with low interest (mean=0.15, standard deviation=0.1), while there is small proportion as enthusiasts or influencers that begin with high interest (mean=0.85, standard deviation=0.08). Influencers are defined as the highest-degree nodes in the network, reflecting the empirical observation that highly connected individuals are disproportionately likely to be an early adopters and opinion leaders. The second attribute is the personal threshold from Gaussian distribution centered on a user-specified mean to capture heterogeneity in how much social proof different agents require before acting.

Each edge in the network is independently labeled as either strong tie or weak tie as friend or a stranger, with probability equal to the friend ratio parameter. Strong ties carry a weight of 1.0 in the influence calculation and weak ties carry a weight of 0.2. At each step, non-terminal agents capture a weighted average of their neighbors' signals, where a neighbor in ACTIVE or SUCCESSFUL state contributes a signal of 1.0, FAILED neighbors contribute -0.2, and others contribute their raw interest level. The agents' interest is updated toward the weighted signal at fixed learning rate of 0.3 and the discrete state is updated accordingly.

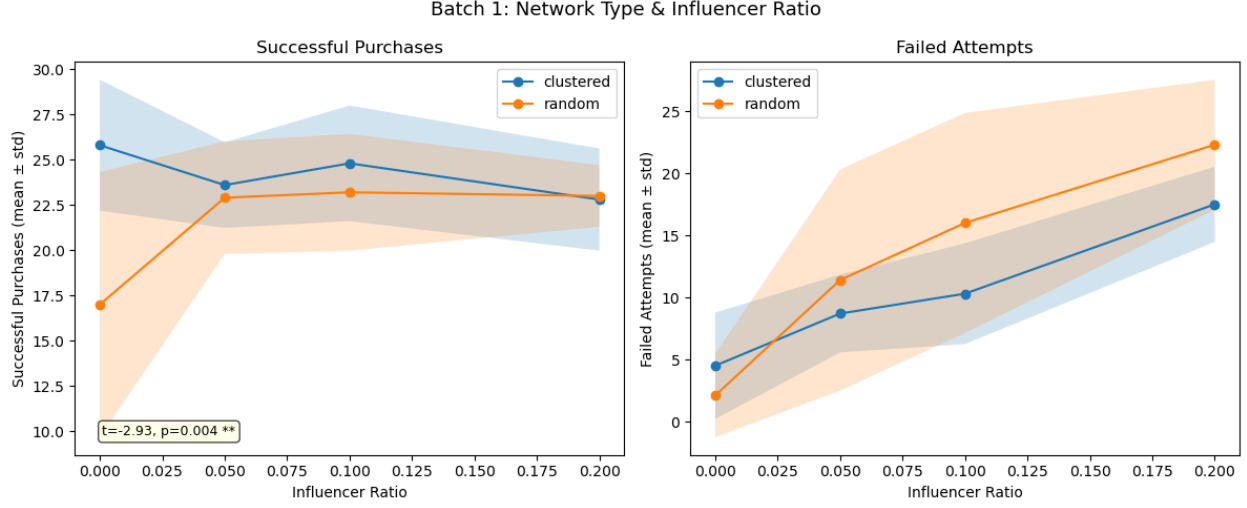
There are five different states in the agent cycle. INDIFFERENT agents have an interest level below 0.3 and have not yet been sufficiently influenced. INTERESTED agents have an interest level between 0.3 and their threshold. ACTIVE agents have crossed their threshold and are attempting to purchase in the current step. SUCCESSFUL agents have obtained the product and no longer act, though they continue to emit a positive signal to neighbors. FAILED agents attempted to purchase but found the supply exhausted; they emit a small negative signal and no longer act. Agents step in a random shuffled order each round, implementing a first-come-first-served allocation that is equivalent to a random queue.

Supply is initialized at a user-specified level and decremented by one each time an agent successfully purchases. At fixed intervals determined by the restock interval parameter, a proportion of the initial supply is restored that up to the original level. The stopping rule happens when no agents remain in the ACTIVE or INTERESTED state, meaning the cascade has fully resolved. Key outputs collected by the DataCollector at each step include the counts of agents in each state, the cascade size (ACTIVE plus SUCCESSFUL), and the remaining supply level.

## Analysis

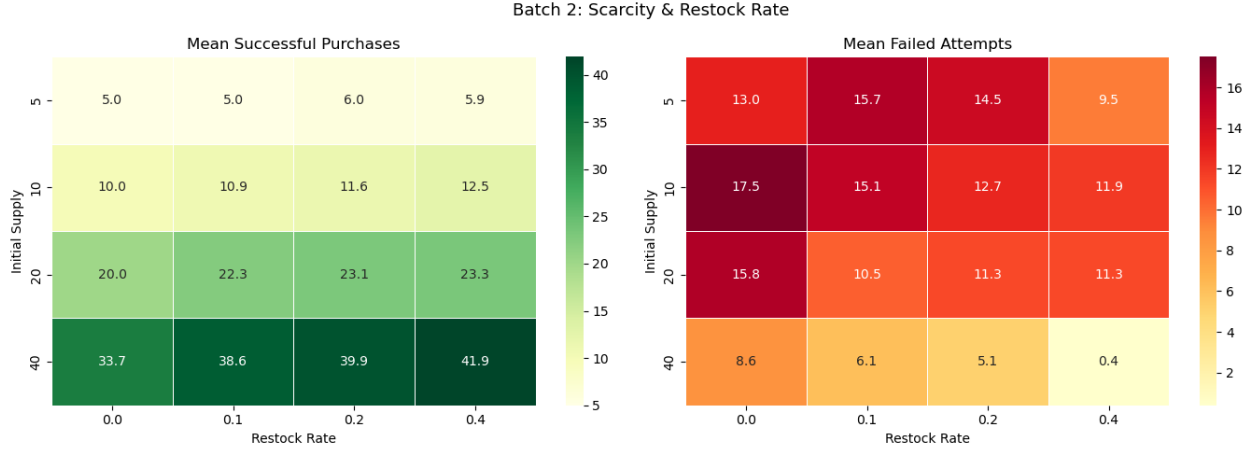
I conducted three batch runs to explore conditions under which demand cascades amplify or collapse. Each parameter combination was repeated ten times with different random seeds.

### Batch Run 1: Network Structure and Influencer Ratio



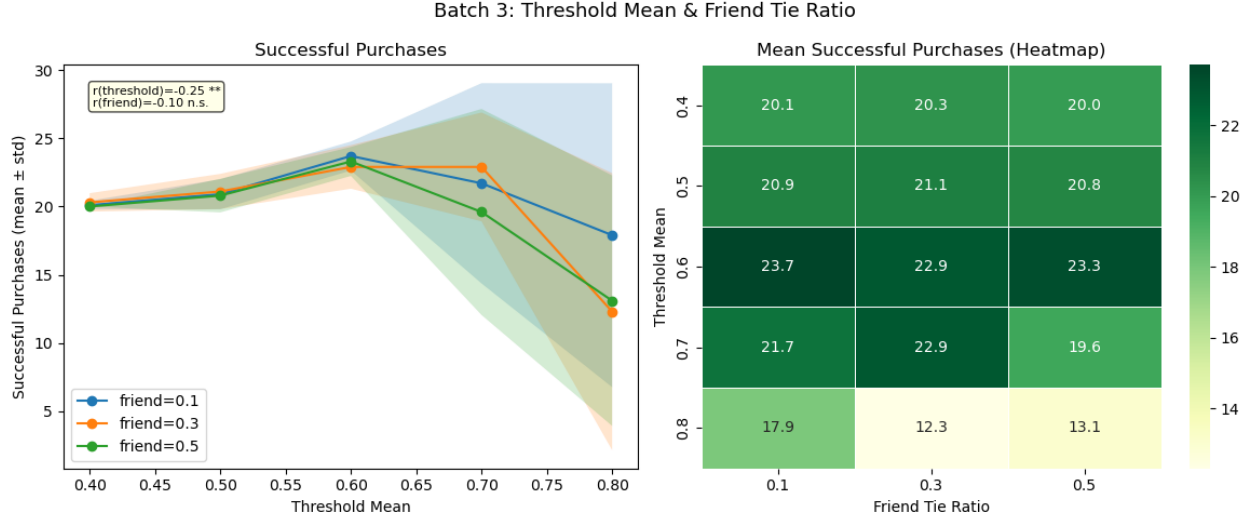
The first batch run varied network type (random versus clustered) and influencer ratio (0%, 5%, 10%, 20%) while holding all other parameters constant. The result shows that clustered networks had significantly more successful purchases overall ( $M = 24.2$ ) compared to random networks ( $M = 21.5, t = -2.93, p = 0.004$ ). This difference was most in the complete absence of influences that without any high-interest seed agents, the clustered network achieved the mean of 25.8 successful purchase compare to only 17 in the random network ( $t = -3.41, p = 0.003$ ). The pattern is consistent with the threshold model logic of Granovetter (1978): early adopters being densely connected to each others in a clustered network, which allows social proof to accumulate locally before spreading outward. Once enough agents within a cluster cross their threshold, the entire cluster tends to activate in sequence. In the random network, the same number of early adopters is spread thinly across the network, and no single neighborhood accumulates enough signal to trigger a local cascade. As the influencer ratio increased, the gap between network types diminished and became non-significant at ratios of 5% and above. This suggests that influencers effectively substitute for structural clustering by broadcasting high-interest signals to many weakly connected agents simultaneously. Notably, the right panel reveals that random networks accumulate substantially more failed purchase attempts at higher influencer ratios, indicating influencer-driven demand in sparse networks outpaces supply availability, producing cascade overshoot rather than cascade completion.

## Batch Run 2: Initial Supply and Restock Rate



The second batch run varied the initial supply of 5, 10, 20, 40 units and the restock rate at 0%, 10%, 20%, 40% of initial supply per interval. The initial supply was the strongest predictor of a successful purchase ( $r = 0.897, p < 0.001$ ), confirming that scarcity is the primary structural constraint on cascade completion. The restock rate did not show a significant overall correlation with successful purchase ( $r = 0.092, p = 0.248$ ), which means the timing and volume of restocking matter less than the absolute quantity of supply available. Pairwise comparison revealed that restock rate had effects only at low supply levels: at supply = 5, restocking at 40% produced modestly more successful purchases than no restocking ( $M = 5.9$  versus  $M = 5$ ;  $t = -2.86, p < 0.01$ ), and the effect was stronger at supply = 10 ( $M = 12.5$  versus  $M = 10$ ;  $t = -4.29, p < 0.001$ ). At higher supply levels of 20 and 40 units, the difference between restock conditions was not statistically significant meaning when supply is already generous relative to demand, restocking only provide little additional benefit. The cascade collapse condition emerged most clearly at the lowest supply level with the initial supply of only 5 units, a mean of just 5 agents succeeded regardless of restock rate, while approximately 12 to 15 agents failed in each condition. This indicates early supply depletion arrested the cascade before social contagion could propagate through the network, agents who had been influenced to act was too late to purchase anything, and their visible failure sent a discouraging signal that dampened the interest of those who had not yet decided.

### Batch Run 3: Threshold Mean and Strong Tie Ratio



The third batch run varied the population threshold mean (0.4 to 0.8) and the proportion of strong ties (10%, 30%, 50%). Friend tie ratio showed no significant correlation with successful purchases ( $r = -0.1, p = 0.223$ ), and pairwise t-tests comparing low and high friend ratios at every threshold level were uniformly non-significant. From the heatmap, successful purchase counts are remarkably stable across all combinations of threshold mean and friend ratio between 0.4 and 0.7, ranging from approximately 20 to 24. This stability finding is surprising because strong tie theory would predict that a higher proportion of friend ties should amplify social influence and facilitate cascade spread. One possible explanation is that in a network with 100 agents and average degree 4, even weak-tie signals accumulate sufficiently across multiple steps to push agents across their thresholds; it renders the strength of any individual tie less consequential. Threshold mean showed a significant negative correlation with successful purchases ( $r = -0.253, p = 0.002$ ), but this effect was largely driven by the highest threshold level: at threshold mean = 0.8, mean successful purchases fell to approximately 13 to 18 across all friend ratio conditions, while all lower threshold levels produced relatively similar outcomes. The results show the threshold ceiling effect, where moderate variation in how much social proof people require does not substantially change aggregate cascade outcomes, but a sufficiently skeptical population can resist the cascade almost entirely.

## Discussion and Conclusion

The model demonstrates that demand in scarcity environments is not a fixed individual attribute but an emergent collective outcome shaped by network structure, tie strength, threshold heterogeneity, and supply dynamics. The most substantively important finding is that clustered networks produce more complete cascades than random networks when influencer density is low, because local clustering allows social proof to accumulate within tight groups before spreading outward. This finding has direct relevance to the design of limited-edition products, where seeding a clustered community with a small number of enthusiastic adopters may be more effective than broadcasting broadly across a sparse random network.

The null finding on friend tie ratio is worth noting as a substantive result. In a sufficiently connected network, the distinction between strong and weak ties matters less for aggregate cascade outcomes than tie strength theories might predict. The effect only appeared at the extremes of the threshold distribution, where very high thresholds suppressed cascade completion regardless of tie composition. Future work could explore sparser networks where the strong or weak tie distinction may prove more consequential.

This model contributes to the existing literature by providing an agent-based operationalization of demand as an endogenous social process, a perspective absent from both the scarcity psychology literature and the informational cascade literature. There were included empirical questions of where real-world social networks sit in the parameter space explored in this model and whether interventions targeting network structure or influencer seeding could either facilitate or disrupt demand cascades in socially relevant contexts.

## External Tool Statement

I found literature through Google Scholar searches using terms related to my idea such as scarcity, threshold models, consumer behavior, and so on. Since I am working on a network model, I structured my code based on network-based examples discussed in class and the official Mesa documentation, including the NetworkGrid API and batch run implementation. I used Claude to clarify how `**kwargs` syntax works in the batch run helper function, to determine that increasing the number of repetitions per parameter combination produces more stable and reliable means.

Link to GitHub Repo: [https://github.com/anniek8/MACS40550\\_ABM](https://github.com/anniek8/MACS40550_ABM)

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